

Updated ethanol spray strength

The ethanol foliar-spray strength in this paper has been standardised across all Pedaver material.

PQNK's recommended strength for an ethanol foliar spray is 1 to 2 percent food-grade ethanol in clean water, about 10 to 20 ml of ethanol per litre. It relieves ambient heat or cold stress, is sprayed in the cooler part of the day, and supports, rather than replaces, the protection of mulch-covered permanent beds. Stronger solutions can scorch leaves.

Where this advisory recommends a 1 percent ethanol solution every 8 to 10 days, use 1 to 2 percent; the interval is unchanged.

The original paper follows unchanged.

PQNK Advisory Note

Frost, Cold Weather & Irrigation Management

(PQNK پقنک – pronounced “picnic” | Paedar Qudratti Nizam Kashatqari: the regenerative & sustainable Pristine Organic Farming System)

The Farmer’s Question (Summary)

During cold and frost conditions (کورا), should irrigation be increased—especially in the morning after frost—as commonly advised in conventional farming? Does water help plants resist frost damage?

PQNK’s Clear Answer (In Short)

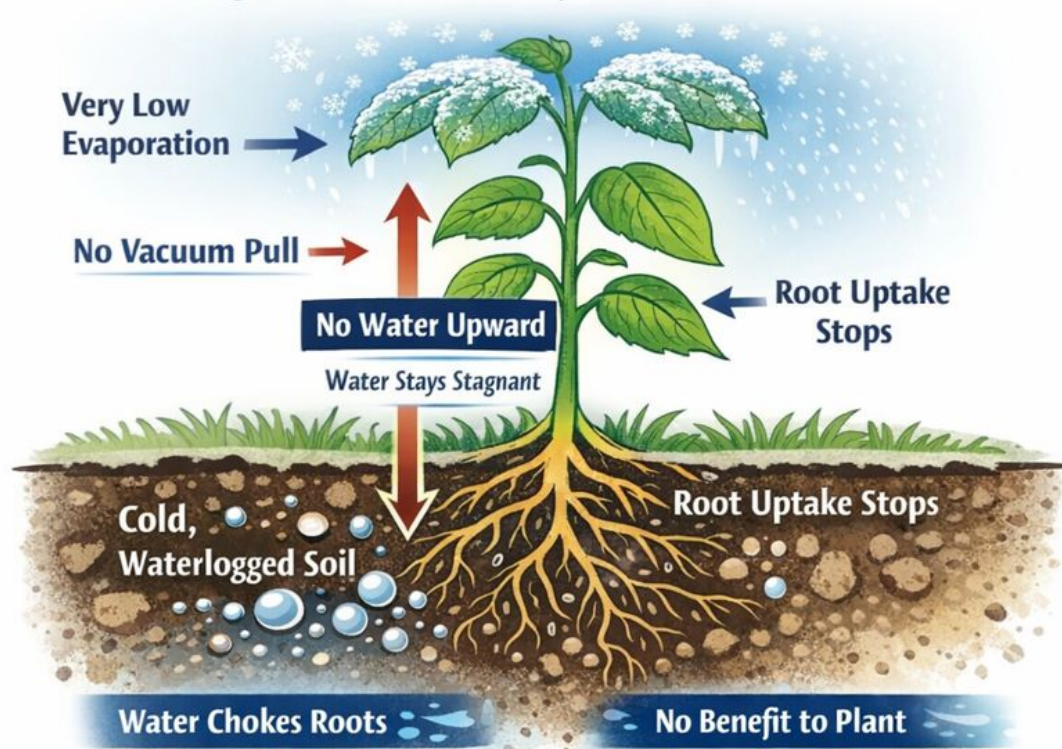
No. Increasing irrigation during cold or frost conditions does not protect plants.

In fact, unnecessary irrigation during cold periods can **harm soil life, reduce aeration, and weaken plants**, whether water comes from canals or tube-wells.

PQNK relies on **soil moisture regulation, soil aeration, organic mulch, and plant immunity**, not emergency watering.

Why Extra Irrigation Fails in Cold Weather

Understanding Water Movement vs Temperature



1. Understanding Cold Stress in Plants (Scientific + Practical)

Cold stress is primarily an **air-temperature and leaf-surface phenomenon**, not a soil-water problem.

- Frost damages plants when **ice crystals form on or inside leaf tissues**, blocking gas exchange.
- Water on or around roots **cannot reverse this process**.
- Plants do **not absorb water on demand**; they absorb water only when leaves are actively transpiring.

In cold conditions:

- Transpiration is **very low**
- Evaporation from leaves is **minimal**
- Water movement inside plants is **naturally slow**

So adding more water **does not push water into the plant**.

2. Two Water Sources – Why Neither Helps During Frost

Farmers often believe warm tube-well water can “save” crops. Let’s examine both sources logically.

(a) Canal Water

- Canal water is at **ambient air temperature**
- During winter, this water is already **cold**
- Applying it:
 - Lowers soil temperature further
 - Fills soil pores with water
 - **Displaces air**, reducing oxygen

(b) Tube-Well (Groundwater)

- Groundwater temperature is usually **18–22°C** (varies by depth & region)
- But once applied:
 - It spreads in a thin layer
 - Loses heat rapidly
 - Reaches soil temperature within minutes

Result:

The warmth is **temporary and meaningless**, but the damage is lasting:

- Soil aeration collapses
- Roots suffocate
- Soil microbes slow down or die
- Anaerobic conditions develop

👉 **There is no lasting thermal benefit**, only biological harm.

3. Why Extra Irrigation Fails – The Core Plant Physiology

Plants uptake water through a **vacuum pull mechanism**:

- Water evaporates from leaves (transpiration)
- This creates a suction force
- Water is pulled upward against gravity

In cold weather:

- Evaporation is extremely low
- Leaves are often stomata-closed
- No vacuum pull exists

So:

- Water applied to soil **cannot enter plants**
- Excess water stays stagnant
- Roots and microbes suffocate

This is true in **both conventional and organic systems**—but PQNK avoids this trap.

4. What PQNK Does Differently (Why It Works)

PQNK prepares crops **before cold arrives**, not after.

PQNK Soil Conditions:

- Well-aggregated soil (crumbly, porous)
- Balanced moisture—not wet, not dry
- High organic matter
- Active soil biology
- Permanent organic mulch cover

Results:

- Soil temperature remains **moderated**
- Sudden temperature shocks are reduced
- Roots stay oxygenated
- Microbial nutrient supply continues
- Plants remain metabolically stable

5. Cold Tolerance of Different Crops (Natural Design)

Not all crops react the same way to cold.

Cold-Tolerant Crops:

- **Wheat**
- **Barley**
- **Mustard**
- **Rapeseed**

These crops contain:

- Natural antifreeze proteins
- Higher soluble sugars
- Strong cell membrane stability

Cold actually **improves** their performance.

Perennial & Tree Crops:

- **Citrus** can tolerate cold when healthy
- Thick cuticle, dense sap, strong roots protect tissues

Dormant Crops & Trees:

- Many species **shed leaves**
- Enter dormancy
- Conserve energy until temperatures rise

This is **survival, not damage**.

6. PQNK-Created Immunity: The Missing Factor

Under PQNK, plants develop **systemic immunity**, not chemical dependency.

This immunity comes from:

- Balanced mineral nutrition (not excess nitrogen)
- On-demand nutrients supplied by microbes
- Strong root systems
- Stable soil moisture
- Aerated soil
- Mulch-regulated temperature

Such plants:

- Have denser sap (less freezing injury)
- Stronger cell walls
- Better stress signaling
- Faster recovery after cold events

Cold becomes a **filter that strengthens plants**, not a threat.

7. Final PQNK Recommendation (Clear & Practical)

- ✓ **Do NOT increase irrigation during frost or cold spells**
- ✓ **Maintain mulch cover**
- ✓ **Trust soil moisture already present**
- ✓ **Allow soil to breathe**
- ✓ **Let plants self-regulate**

One Line to Remember:

Cold damages weak systems, not strong ones. PQNK builds strength before stress arrives.

Special PQNK Advisory for Mild-Weather & Transitional-Season Crops

(Early Spring & Late Autumn Conditions)

In regions like the **Punjab plains of the Indian subcontinent**, spring and autumn are **very short transition periods**. Crops planted during these windows—such as **potato, maize (corn), early vegetables, and late vegetables**—often face **physiological stress** due to rapid temperature swings rather than true water shortage.

1. The Core Problem in Transitional Seasons

- **Early spring:**
 - Low night temperatures delay germination
 - Soil moisture may be present, but **metabolic activity is slow**
- **Late autumn:**
 - Soil remains warm while air temperature drops rapidly
 - This creates **root–shoot imbalance** and stress in conventional systems

In both cases, **extra irrigation does not solve the problem**, because:

- Water uptake depends on **leaf evaporation (transpiration)**
- In cool or unstable temperatures, evaporation is minimal
- Excess water only **reduces soil aeration** and stresses roots

2. PQNK Recommendation: 1% Ethanol Spray (Biologically Compatible Intervention)

For **mild-weather crops** and **early/late season vegetables**, PQNK recommends:

Spraying a 1% ethanol solution every 8–10 days

Why ethanol works (PQNK scientific rationale):

- Ethanol acts as a **natural cryo-protectant**, similar to compounds plants produce internally
- It:
 - Lowers cellular freezing stress
 - Stabilizes cell membranes
 - Improves enzymatic activity during temperature fluctuations
- It enhances **stress tolerance** without disturbing soil life
- It mimics **natural antifreeze behavior** seen in cold-tolerant plants

PQNK avoids chemical inputs but **accepts biologically compatible tools** when they align with natural plant physiology.

This practice is **not a fertilizer, not a pesticide, and not a stimulant**—it is a **temporary adaptive support** during climatic transitions.

3. About Midnight Tube-Well Irrigation & Frost

A common question is whether applying **tube-well water at midnight** helps protect crops from frost.

Scientific clarification:

- Groundwater usually emerges **slightly warmer** ($\approx 18\text{--}22^\circ\text{C}$ depending on depth)
- However:
 - Once applied, water **rapidly cools** to ambient temperature
 - It fills soil pores, **displacing air**
 - This **reduces soil respiration** and harms roots and soil life

While **temporary surface warmth** may slightly delay frost for a few hours, the **net effect is negative** in most cases:

- Roots become oxygen-starved
- Soil temperature drops again before sunrise
- Plant stress increases rather than decreases

PQNK position:

- **Do not rely on midnight irrigation** as a frost-control strategy
- A **mulched, aerated, biologically active soil** provides far better frost buffering
- Soil insulation + plant immunity is superior to water flooding

4. The PQNK Bottom Line for Transitional Crops

- Frost and cold stress are **energy and physiology problems**, not water problems
- PQNK solves these through:
 - Regulated soil moisture (never saturation)
 - Organic mulch for temperature buffering
 - Active soil biology
 - Plant-level immunity
- **Ethanol spray** is recommended only where:
 - Crops are outside their ideal season
 - Temperature fluctuations are unavoidable
 - Natural immunity needs temporary support

Final Farmer Takeaway (Simple Logic)

 **Cold slows evaporation → no water pull → irrigation fails**

 **Healthy soil + mulch + biology = natural protection**

 **Ethanol helps plants adapt, not depend**

This approach keeps crops **alive, balanced, and productive**—without breaking natural laws.

Footnote: PQNK (pronounced 'picnic')—the regenerative & sustainable Pristine Organic Farming System—is grounded in Natural Ecosystem Science. This science employs perennial biodiversity, protects soil with organic mulch, and avoids practices like water inundation, tillage disturbance, and leaving soil bare.

It is the process that matters most. PQNK is a pristine production system designed not merely for yield, but for optimal, balanced output—where quality, sustainability, and ecological harmony emerge from a self-regulated, zero-external-input methodology.

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